

Multiple Choice Questions

1. In C++, a destructor is automatically called:
 - A) When the object is manually deleted.
 - B) When the object goes out of scope.
 - C) When the `Dispose` method is called.
 - D) When the program starts.
2. The purpose of the destructor in the `Stack` class (C++) is to:
 - A) Initialize the stack.
 - B) Clear all elements in the stack.
 - C) Release the memory allocated for the array.
 - D) Reallocate the array to a new size.
3. In C#, which method is used to clean up unmanaged resources?
 - A) `CleanUp()`
 - B) `Dispose()`
 - C) `Clear()`
 - D) `Delete()`
4. The `Finalize` method in C# is declared using which of the following keywords?
 - A) `Dispose`
 - B) `~` (Tilde)
 - C) `Destructor`
 - D) `CleanUp`
5. The `using` block in C# ensures that:
 - A) The object is collected by the garbage collector immediately.
 - B) The object is manually disposed of.
 - C) `Dispose` is called on the object automatically.
 - D) The object cannot be used outside the block.
6. What happens when `GC.SuppressFinalize(this)` is called?
 - A) The object is immediately destroyed.
 - B) The finalizer is skipped for the object.
 - C) The object is automatically disposed of.
 - D) The object is locked in memory.
7. In the `MyFileReader` class, the destructor is responsible for:
 - A) Creating a new file.
 - B) Closing the file handle.
 - C) Reading the file.
 - D) Opening the file again.
8. A class that implements `IDisposable` in C# must:

- A) Provide a destructor.
 - B) Provide a `Dispose` method.
 - C) Use a `try...catch` block.
 - D) Be a sealed class.
9. In the `MySuperFileReader` class, which of the following is a managed resource?
- A) File handle (`IntPtr`)
 - B) The operating system's file handle
 - C) The `StreamWriter` object
 - D) The native DLL function
10. The `isDisposed` flag in the `MySuperFileReader` class is used to:
- A) Track the number of times the object is disposed.
 - B) Prevent multiple disposal of the object.
 - C) Store the object's state.
 - D) Count the number of open files.
11. In the `MySuperFileReader` class, which method is directly called by the destructor?
- A) `Dispose(true)`
 - B) `Dispose(false)`
 - C) `SuppressFinalize()`
 - D) `Close()`
12. What is the main purpose of the `Dispose` method in `MySuperFileReader`?
- A) To read the file.
 - B) To free both managed and unmanaged resources.
 - C) To reallocate the file handle.
 - D) To open a new file.
13. What does the `try...finally` block in the `StreamReader` example ensure?
- A) The object is re-used.
 - B) The object is disposed of even if an exception occurs.
 - C) The object is never disposed.
 - D) The code runs faster.
14. Which of the following is true about unmanaged resources?
- A) They are directly managed by the .NET runtime.
 - B) They are automatically released by garbage collection.
 - C) They must be manually released using `Dispose`.
 - D) They do not exist in C#.
15. In the `MyFileReader` class, which function is used to close the file handle?
- A) `Dispose`
 - B) `GC.Collect`

C) `FileFunctions.CloseHandle`

D) `Finalize`

16. What is the role of the finalizer in the `MySuperFileReader` class?

A) To immediately dispose of the object.

B) To ensure unmanaged resources are released if `Dispose` is not called.

C) To prevent the object from being garbage collected.

D) To reallocate memory.

17. The `StreamReader` object is properly disposed when:

A) It is declared in a `using` block.

B) It is declared in a `try...catch` block.

C) It is declared as a static object.

D) It is declared as a global variable.

18. In the `MySuperFileReader` class, the `dispose` method is:

A) Automatically called by the garbage collector.

B) Only called in the finalizer.

C) Called by both `Dispose` and the finalizer.

D) Called when the object is created.

19. Which of the following is true about `GC.Collect()`?

A) It guarantees immediate object destruction.

B) It triggers garbage collection but does not guarantee finalizer execution.

C) It is automatically called for all `IDisposable` objects.

D) It should always be used for resource management.

20. In the `MySuperFileReader` class, calling `Dispose` directly:

A) Also calls the finalizer.

B) Prevents the finalizer from being called.

C) Calls the `Dispose` method of the base class.

D) Has no impact on the finalizer.

Answer Key

1. B
2. C
3. B
4. B
5. C
6. B
7. B
8. B
9. C
10. B
11. B
12. B
13. B
14. C
15. C
16. B
17. A
18. C
19. B
20. B

Multiple-Answer Questions (Select All That Apply)

1. Which of the following are characteristics of a destructor in C++?
 - ☐ It is called manually by the programmer.
 - ☐ It has the same name as the class but starts with a tilde (~).
 - ☐ It cannot be overloaded.
 - ☐ It is automatically called when an object goes out of scope.
2. In the `Stack` class (C++), which of the following are true?
 - ☐ The array is dynamically allocated using `new`.
 - ☐ The array is released using `delete`.
 - ☐ The destructor is responsible for freeing memory.
 - ☐ The destructor is named `Dispose`.
3. Which of the following methods are part of resource management in C#?
 - ☐ `Dispose`
 - ☐ `Finalize`
 - ☐ `GC.Collect`
 - ☐ `DisposeAll`
4. Which of the following statements about the `Finalize` method in C# are true?
 - ☐ It is implicitly called by the garbage collector.
 - ☐ It must be manually called by the programmer.
 - ☐ It is declared using the `~` symbol in C#.
 - ☐ It is protected and can be overridden.
5. Which of the following are true about the `Dispose` method?
 - ☐ It is a part of the `IDisposable` interface.
 - ☐ It is called automatically when an object goes out of scope.
 - ☐ It is explicitly called to release resources.
 - ☐ It is used to clean up unmanaged resources.
6. Which statements about the `using` block in C# are correct?
 - ☐ It can only be used with classes that implement `IDisposable`.
 - ☐ It ensures `Dispose` is called on the object.
 - ☐ It can handle multiple objects at once.
 - ☐ It prevents exceptions from occurring.
7. Which of the following are unmanaged resources?
 - ☐ File handles in Windows API
 - ☐ Database connections managed by ADO.NET
 - ☐ Memory allocated with `new` in C++
 - ☐ A `StreamWriter` object in C#
8. What does `GC.SuppressFinalize(this)` do?

- ☐ Prevents the garbage collector from collecting the object.
 - ☐ Prevents the finalizer of the object from being called.
 - ☐ Makes the object eligible for immediate collection.
 - ☐ Is automatically called in all `IDisposable` classes.
9. In the `MyFileReader` class, which of the following are true?
- ☐ It uses a native DLL function to open the file.
 - ☐ The file handle is managed by the .NET runtime.
 - ☐ The file handle is released in the finalizer.
 - ☐ It uses a managed `StreamReader` for file reading.
10. Which of the following are true for the `MySuperFileReader` class?
- ☐ It implements `IDisposable`.
 - ☐ It has both managed and unmanaged resources.
 - ☐ It calls `Dispose` in the finalizer.
 - ☐ It uses a `try...finally` pattern for cleanup.
11. What are the benefits of using `try...finally` with `Dispose`?
- ☐ Ensures cleanup even in case of exceptions.
 - ☐ Makes code faster.
 - ☐ Guarantees that `Dispose` is only called once.
 - ☐ Simplifies resource management.
12. Which of the following are true about the `dispose` method in `MySuperFileReader`?
- ☐ It is private.
 - ☐ It handles both managed and unmanaged resource cleanup.
 - ☐ It is called from both `Dispose` and the finalizer.
 - ☐ It must be manually called by the user.
13. Which of the following are considered managed resources?
- ☐ A `StreamWriter` in C#
 - ☐ An integer array in C#
 - ☐ A file handle in the Windows API
 - ☐ A `List<string>` in C#
14. What are the key characteristics of a class implementing `IDisposable`?
- ☐ It must provide a `Dispose` method.
 - ☐ It must have a finalizer.
 - ☐ It must clean up both managed and unmanaged resources.
 - ☐ It must always use `GC.SuppressFinalize(this)`.
15. Which of the following are true about `System.GC.Collect()`?
- ☐ It forces immediate garbage collection.
 - ☐ It guarantees that all finalizers will be called.

- ☐ It is generally recommended for regular use in applications.
- ☐ It can negatively impact performance.

16. Which of the following are true about `StreamReader` in C#?

- ☐ It must always be disposed of to free resources.
- ☐ It is a managed resource.
- ☐ It can be used with a `using` block.
- ☐ It automatically calls `Dispose` when it goes out of scope.

17. Which of the following are true for a class with unmanaged resources?

- ☐ It must implement `IDisposable`.
- ☐ It must provide a finalizer.
- ☐ It must always use `GC.Collect()`.
- ☐ It should ensure proper disposal of unmanaged resources.

18. In the `MySuperFileReader` class, which are true about the `isDisposed` flag?

- ☐ It prevents multiple disposal.
- ☐ It is set to `true` after `Dispose` is called.
- ☐ It is a required part of `IDisposable`.
- ☐ It ensures that unmanaged resources are released only once.

19. Which of the following are true about multiple `using` statements?

- ☐ They can be nested.
- ☐ They can be written in a single line.
- ☐ Each `using` object is disposed of independently.
- ☐ Only the first object is disposed of.

20. Which of the following are true about unmanaged resources?

- ☐ They are managed by the .NET garbage collector.
- ☐ They must be explicitly released by the developer.
- ☐ They include file handles, database connections, and native memory.
- ☐ They are automatically released when an object is destroyed.

Answer Key

1. ☒, ☒, ☒, ☒
2. ☒, ☐, ☒, ☐
3. ☒, ☒, ☒, ☐
4. ☒, ☐, ☐, ☒
5. ☒, ☐, ☒, ☒
6. ☒, ☒, ☒, ☐
7. ☒, ☒, ☐, ☐
8. ☐, ☒, ☐, ☐
9. ☒, ☐, ☒, ☐
10. ☒, ☒, ☐, ☐
11. ☒, ☐, ☐, ☒
12. ☒, ☒, ☒, ☐
13. ☒, ☒, ☐, ☒
14. ☒, ☐, ☒, ☐
15. ☒, ☐, ☐, ☒
16. ☒, ☒, ☒, ☐
17. ☒, ☒, ☐, ☒
18. ☒, ☒, ☐, ☒
19. ☒, ☒, ☒, ☐
20. ☐, ☒, ☒, ☐

True or False Questions

1. In C++, a destructor is automatically called when an object goes out of scope.
2. In the provided `Stack` class (C++), the destructor correctly releases the allocated array using `delete[]`.
3. The C# `Stack` class requires a `delete` statement in the destructor to clean up the array.
4. A `Finalize` method in C# is automatically called for every class without any need for implementation.
5. The `Dispose` method is a standard part of every class in C#.
6. The `using` block in C# is only useful for unmanaged resources.
7. In C#, the `GC.SuppressFinalize(this)` prevents the finalizer from being called on the object.
8. The `MyFileReader` class uses an unmanaged file handle, which must be manually closed.
9. The `MySuperFileReader` class manages both managed and unmanaged resources.
10. In the `MySuperFileReader` class, the destructor calls the `Dispose` method directly.
11. If the `Dispose` method is called explicitly, the destructor is still called automatically.
12. The `try...finally` pattern ensures that a resource is released even if an exception occurs.
13. The `Dispose` method in `MySuperFileReader` calls `GC.SuppressFinalize(this)` to avoid a double release of resources.
14. An unmanaged resource in C# is one that is directly managed by the .NET runtime.
15. The `using` block in C# guarantees that `Dispose` will be called, even if an exception is thrown.
16. The `StreamReader` class in C# automatically calls `Dispose` when it goes out of scope.
17. A `Dispose` method can be called multiple times without any error in `MySuperFileReader`.
18. In `MySuperFileReader`, the `dispose` method is private and can only be called internally.
19. Calling `System.GC.Collect()` guarantees that all objects will be immediately destroyed.
20. The `MySuperFileReader` class implements `IDisposable` to ensure proper resource management.

Answer Key

1. **True**
2. **True**
3. **False**
4. **False**
5. **False**
6. **False**
7. **True**
8. **True**
9. **True**
10. **False**
11. **False**
12. **True**
13. **True**
14. **False**
15. **True**
16. **False**
17. **True**
18. **True**
19. **False**
20. **True**

Complete the Sentence Questions (One-Word Answer Only)

1. The special method used to clean up resources in C++ is called a _____.
2. In C++, the keyword used to release dynamically allocated memory is _____.
3. In C#, the method used to release unmanaged resources is called _____.
4. The using block in C# automatically calls the _____ method on an object.
5. A method in C# that is similar to a destructor but is automatically called by the garbage collector is _____.
6. The Dispose method is defined in the _____ interface.
7. The .NET method that forces garbage collection is _____.
8. In the MySuperFileReader class, the file handle is stored as an _____.
9. The dispose method in MySuperFileReader is called by the finalizer with the argument _____.
10. The process of preventing the finalizer from being called is done using _____.
11. The try...finally pattern ensures that _____ is always called, even during exceptions.
12. A resource that is not managed by the .NET runtime is called an _____ resource.
13. In the Stack class (C++), the destructor uses _____ to release the array.
14. The base class method for cleanup in C# is called _____.
15. In C#, the keyword used to create an instance of a class is _____.
16. The method FileFunctions.CloseHandle is used to release a _____ resource.
17. The isDisposed flag is used to prevent multiple _____ of an object.
18. The class MySuperFileReader uses a _____ block to ensure disposal of the StreamWriter.
19. The method GC.SuppressFinalize is called inside the _____ method.
20. A method that must be called explicitly to release resources is called _____.

Answer Key

1. **destructor**
2. **delete**
3. **Dispose**
4. **Dispose**
5. **Finalize**
6. **IDisposable**
7. **GC.Collect**
8. **IntPtr**
9. **false**
10. **GC.SuppressFinalize**
11. **Dispose**
12. **unmanaged**
13. **delete[]**
14. **Finalize**
15. **new**
16. **unmanaged**
17. **disposal**
18. **using**
19. **Dispose**
20. **Dispose**